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ORIGINAL PAPER

A glassy carbon electrode modified with in-situ generated chromium-loaded CdS nanoprobes and heparin for ultrasensitive electrochemiluminescent determination of thrombin

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Abstract We describe the in-situ electrochemical synthesis of a chromium-loaded CdS nanoprobe (CdS-Cr) for use in ultrasensitive electrochemiluminescence (ECL) detection of thrombin. A glassy carbon electrode was covered with (a) a film of electro-polymerized polyaniline nanofibers (PANI-NF), (b) the in-situ synthesized CdS-Cr nanoprobe, (c) heparin, and (d) a coating of BSA to prevent unspecific adsorption. Atomic force microscopy and scanning electron microscopy images showed the size of nanoprobe to have increased to an average size of 50±5 nm, which was larger than pure CdS nanocrystals (NCs) with their typical size of 10±2 nm. The modified glassy carbon electrode was electrochemically characterized by cyclic voltammetry and impedance spectroscopy. The thrombin-heparin interaction served as a new recognition tool for thrombin and represented an attractive alternative to respective aptamers. About 6-fold enhancement of ECL intensity (compared to pure CdS NCs) was observed in presence of persulfate. This ECL assay has a wide dynamic range (from 10 fM to 100 pM concentrations of thrombin), and a lower detection limit of 6.8 fM at 3σ. The technique for in-situ electrochemical preparation of the nanoprobe is simple, facile, and yields a sensor surface with favorable space structure, positive charge and stability.

Keywords Chromium-loaded CdS nanoprobe · Electrochemiluminescence · Thrombin · Heparin · Polyaniline nanofibers · Cyclic voltammetry · Impedance spectroscopy

Introduction

Electrochemiluminescence (ECL) has been a very powerful analytical technique due to its high sensitivity, simplified optical setup, low background signal, good spatial controllability [1]. Due to the unique quantum size dependent electrochemical properties of semiconductor nanocrystals (NCs) and controllable ECL merits [2], NCs, have become fascinating luminophores for the fabrication of biosensors. The prepared ECL biosensors have been widely applied in immunoassay of protein, DNA detection, carbohydrate analysis and cytosensing [3–9]. However, compared with conventional luminescent reagents like luminal or $Ru(bpy)_3^{2+}$, NCs usually suffer from relatively weaker ECL emissions. Thus, finding a way to obtain high ECL efficiency of NCs for bioanalysis is the constant driving force of this area. Many previous works have demonstrated that ECL emissions of NCs are highly dependent on their surface states, which can be changed by coupling with other nonmaterial or doping some metal ions [10–17]. Deng et al. and Wang et al. reported that the doping of Mn^{2+} or Eu^{3+} ions in the CdS NCs surface alters the surface of CdS NCs and builds a new surface state-Mn^{2+} or Eu^{3+} complex, also produces a maximum of 4-fold enhancement in ECL intensity [16, 17]. Kumar et al. reported that CdS nanoparticles doped with 2 and 4 at. % of Cr can be synthesized in aqueous solution by chemical co-precipitation by using poly(vinyl pyrrolidone) as a stabilizer [18]. Although metal ion-doped water dispersible semiconductors can amplify the ECL signal of nanoprobes, they are synthesized not via in-situ system, and the process of

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preparing is complicated and time-consuming. Therefore, the development of a method for simple, facile and in-situ synthesis is of great significance to develop the ECL system.

Electrochemical synthesis is an attractive method for the synthesis of semiconductor thin films and nanostructures due to it is convenient, quick and parameter controllable [19]. Recently, Xu et al. synthesized the multi-walled carbon nanotubes-CdS nanocomposite by using electrochemical synthesis, and it showed sensitive ECL responses compared with that of the MWCNT-CdS composite obtained under simple ultrasonic mixing [11]. Thus it provides an idea for us to synthesize chromium-loaded ECL nanoprobe via in-situ system, which were expected to amplify the ECL signal of nanoprobes and improve the detection sensitivity.

Thrombin is a protease which plays important role in many life processes, such as blood coagulation, incrustation and inflammation. It also relates to many diseases including Alzheimer's disease and thromboembolic disease [20]. As a recognition element of thrombin, aptamer is single stranded DNA or RNA oligonucleotides selected from random sequence nucleic acid libraries through an in vitro selection process by exponential enrichment (SELEX) [21]. Aptamer-based electrochemical biosensor has been widely used to detection thrombin [22–24]. Han et al. showed perylene-3, 4, 9, 10-tetracarboxylic-toluidine blue (PTC-Tb) probe-based aptasensor for electrochemical and ECL detection of thrombin [22]. Jie et al. demonstrated Fe_3O_4@CdSe composite quantum dot-based ECL aptasensor for the detection of thrombin [23]. Li et al. reported an ECL aptasensor for the selective detection of thrombin by using luminol and its analogs functionalized AuNPs [24]. These studies were used to detect thrombin by using the specific recognition of thrombin with its aptamer. However, to the best of our knowledge, heparin-based assay for thrombin is rarely studied. Heparin is a highly sulfated acidic glycosaminoglycan, which contains functional groups of sulfate ($-OSO_3^-$, $-NHSO_3^-$) and carboxylate ($-COO^-$) [25]. Huntington reported that the heparin specifically binds anion-binding exposits II (heparin soleplate on endothelial cells or chondroitin soleplate on some molecules of thrombomodulin) of thrombin [26]. Moreover, compared to aptamer, heparin is cheap, readily available and similar identification capability for thrombin. Also the aptamer requires chemical modification, and the sensitivity requires improved. Thus, as a new material instead of aptamer, it is of great significance to detect thrombin by using the specific recognition of thrombin with heparin in bioanalytical chemistry.

Herein, we present the in-situ electrochemical synthesis of a chromium-loaded CdS (CdS-Cr) nanoprobe for the ultrasensitive electrochemiluminescence (ECL) determination of thrombin. Due to large surface area, high conductivity, many microgaps existing and positive charges, polyaniline nanofibers (PANI-NF) has caused widespread concern in the construction of biosensor [27]. Polyaniline nanofibers (PANI-NF) films were electropolymerized onto the surface of bare glassy carbon electrode (GCE), which was used as a template. Then chromium-loaded CdS (CdS-Cr) nanoprobes were further formed on PANI-NF by in-situ electrochemical synthesis. Heparin was directly immobilized onto the GCE electrode modified by CdS-Cr/PANI-NF. The specific interaction of thrombin with heparin resulted in the decrease of ECL intensity, which could be used to detect the thrombin. The morphology and size of CdS-Cr nanoprobes were revealed by atomic force microscopy and scanning electron microscopy images. The analytical performance of ECL biosensor and the applicability in human serum were researched.

Experimental

Reagents and apparatus

Human α-thrombin (500 U), hemoglobin (Hb), immunoglobulin G (IgG) and bovine serum albumin (BSA) were purchased from Sigma Chemical Co. (St. Louis, MO, http://www.sigmaaldrich.com). Cadmium chloride ($CdCl_2 \cdot 2.5H_2O$), cadmium nitrate ($Cr(NO_3)_3$), heparin sodium (160 IU $mg \cdot L^{-1}$), and sodium sulfide (Na_2S) were purchased from Sino Pharm Chemical Reagent Co. Ltd, (Shanghai, China, http://www.sinoreagent.com.cn). Unless otherwise stated, the chemicals in this work were all analytical grade reagents; they were used as received without further purification. Double-distilled water was used throughout the experiments. Phosphate buffer (0.1 M, pH 7.4) contained 0.1 M Na_2HPO_4, 0.1 M $KHPO_4$ and 0.1 M KCl.

Electrochemical experiments were performed on a CHI-660D electrochemical workstation (Shanghai Chenhua, http://www.chinstr.com); and electrochemiluminescence curves were recorded using a Model MPI-E from ECL Analyzer Systems (Xi'an Remix Electronic Science & Technology Co. Ltd, http://www.chinaremex.com). All the electrochemical and ECL measurements were carried out on a standard three-electrode system: the working electrode–either the untreated or modified GCE (d=3 mm); the counter electrode–a platinum wire, and the reference electrode–an Ag/AgCl (sat. NaCl). The morphologies of the nanoprobes was visually characterized by scanning electron microscopy (Philips Electron Optics, Netherland, JEOL-JSM-6700F, http://www.fei.com) and Auto Probe CP Research atomic force microscopy (Germany, bruker, http://bruckgmbh.cn.china.cn) in the tapping mode.

Preparation of the chromium-loaded CdS (CdS-Cr) nanoprobes

The glassy carbon electrode (GCE) was pretreated before modification by polishing its surface with aqueous slurries

of alumina powders (0.3 and 0.05 μm Al_2O_3) on a polishing silk. Then the electrode was sonicated in nitric acid (1:1, *v/v*), ethanol and double-distilled water, consecutively.

The electro-polymerization of PANI-NF was carried out by immersing the GCE in an unstirred, fresh solution containing 50 μL distilled aniline, 0.5 mL 1.0 M H_2SO_4, 9 mL double-distilled water, and performing at a current of 120 μA for 300 s. After electro-polymerization, the modified electrode was thoroughly rinsed with water to remove any ions, which might had been weakly, physically absorbed, and then it was dried in air. In-situ synthesis of the CdS-Cr nanoprobe was performed on the surface of PANI-NF-modified GCE. Thus, the modified GCE was dipped into a 9 mL solution containing 0.1 M $CdCl_2$, 0.1 M Na_2S, which had been pre-purged with pure nitrogen for 15 min to expel the dissolved oxygen. Then, the 320 μA current was applied for 300 s. Following this step, 0.0800 g $Cr(NO_3)_3$ was added to the above solution, and 320 μA current was applied for next 300 s. Finally, the modified electrode was rinsed several times with water and dried in air.

Fabrication of the biosensor

PANI-NF film was modified onto the surface of electrode via electro-polymerization technique, and then chromium-loaded CdS nanoprobes (CdS-Cr) were further synthesized onto the PANI-NF film. By using electrostatic interactions between heparin and PANI-NF, and metal ions complexation between heparin and Cr^{3+} or Cd^{2+}, the modified electrode was incubated in PBS buffer solution (0.1 M, pH 7.4) containing 0.8 mg·mL^{-1} heparin for about 2 h to bind heparin. Then, 1 wt % bovine serum albumin solution was used to block the nonspecific binding sites for 1 h. Finally, the ECL biosensor was rinsed with phosphate buffer (0.1 M, pH 7.4), dried and stored at 4 °C for use. The fabrication procedure of ECL biosensor is illustrated stepwise in as shown in Scheme 1.

Analytical procedures

For performing ECL measurement, the electrolytic cell was placed directly in front of a photomultiplier (PMT, operated at 750 V), the scan range was from 0 to −1.8 V, and the linear scan potential was applied with a scan rate of 100 mV·s^{-1}. Sample solutions containing various concentrations of thrombin were prepared in phosphate buffer (0.1 M, pH 7.4). In a typical test, the assembled ECL biosensor was incubated in 100 μL of sample solution for 120 min at 37 °C, followed by washing with phosphate buffer (0.1 M, pH 7.4) to remove unbound thrombin. Before and after the incubation, the ECL responses of the electrode were both recorded in phosphate buffer (0.1 M, pH 7.4) containing 0.1 M $K_2S_2O_8$ as coreactant. The ECL signals related to the thrombin concentrations were measured.

Real samples analysis

The human serum from healthy volunteers were diluted 10-fold in Tris–HCl buffer (20 mM Tris–HCl, 100 mM NaCl, 25 mM KCl, and 2 mM $MgCl_2$, pH 7.4). Three concentrations of thrombin (5.0, 10.0, and 100 pM) were detected in the 10-fold diluted human sera.

Result and discussion

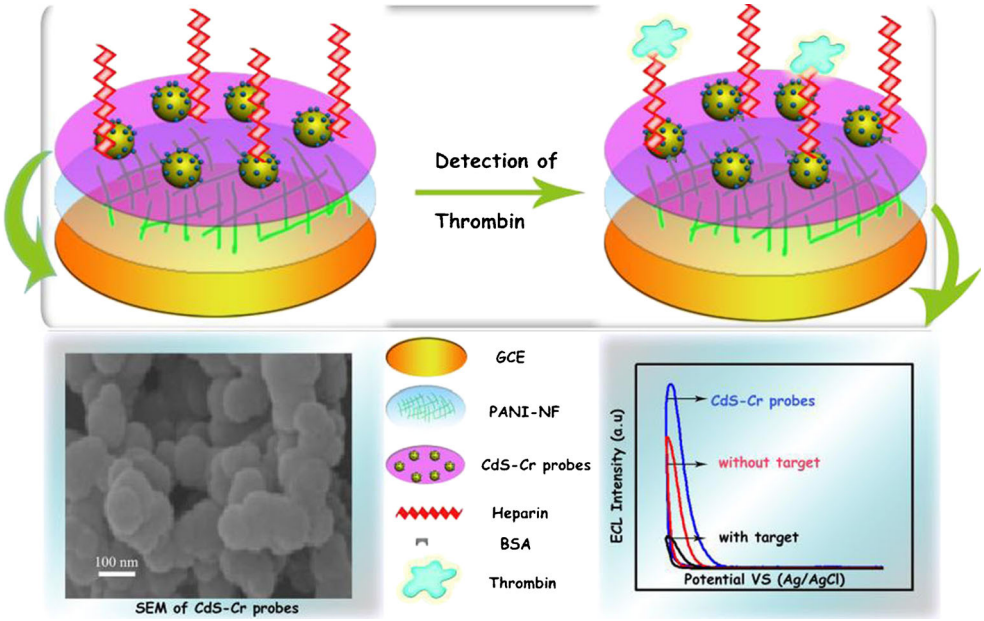
Characterization of the chromium-loaded CdS (CdS-Cr) nanoprobes

High resolution scanning electron microscopy (SEM) and atomic force microscope (AFM) were employed to reveal the morphology of PNAI-NF/GCE, CdS NCs/ PNAI-NF/GCE and CdS-Cr/ PNAI-NF/GCE electrodes. The electro-polymerized polyaniline nanofibers (PANI-NF) showed fiber structural, much greater exposed surface area, and an average size of 20±3 nm (Fig. 1a). The pure CdS NCs, which were in-situ synthesized on the PANI-NF thins, exhibited uniform morphology with an average size of 10±2 nm, and it evenly dispersed on the surface of PANI-NF, as shown in Fig. 1b. However, by co-electrodeposition solutions containing 0.1 M Cd^{2+}+0.025 M Cr^{3+}+0.1 M Na_2S on PANI-NF film modified electrode, the chromium-loaded CdS nanoprobes (CdS-Cr) was formed. Also, the size of nanoprobes increased to 50± 5 nm as shown in Fig. 1c.

The surface topographies of CdS NCs/PANI-NF and CdS-Cr/PANI-NF were also observed by AFM, and the image was shown in Fig. 1d and e, respectively. Network-like structures were found in Fig. 1d, also a number of particles were located on the surface of PANI-NF, which indicated the formation of CdS NCs/PANI-NF. However, After CdS-Cr nanoprobes were modified on PANI-NF film as shown in Fig. 1e, obviously, the size of CdS-Cr nanoprobe increased, and the surface of CdS-Cr/PANI-NF became dense and compact. The results were consistent with the above SEM images.

Electrochemical characterizations of the biosensor

Cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) were used to verify the assembly processes of the modified electrodes (Fig. 2). A potential versus current plot (curve a, Fig. 2a) showed that the $Fe(CN)_6^{4-/3-}$ redox couple produced two well-defined reversible peaks at the GCE. When PANI-NF film were electropolymerized onto the surface of bare GCE, the intensity of both the cathodic and anodic peak current profiles increased substantially (curve b); this observation may be attributed to the attraction between the positive charges on the surface of the PANI-NF film and $Fe(CN)_6^{4-/3-}$. Then the CdS-Cr nanoprobes were formed on



Scheme 1 Diagrammatic representation of the preparation process of the biosensor and the electrochemiluminescence detection of thrombin

them, the peak currents of the electrode further increased (curve c), which indicated that the CdS-Cr nanoprobes enhanced the electron transfer. When heparin and bovine serum albumin (BSA) were immobilized onto the above electrode respectively, an obviously decrease in the amperometric signal was found (curve d and e). Then the peak currents further decreased, and the gap between the anodic and cathodic peaks became wider after subsequent specific recognition of thrombin with heparin (curve f). The phenomenon resulted from the electron inert feature of heparin and thrombin, which blocked

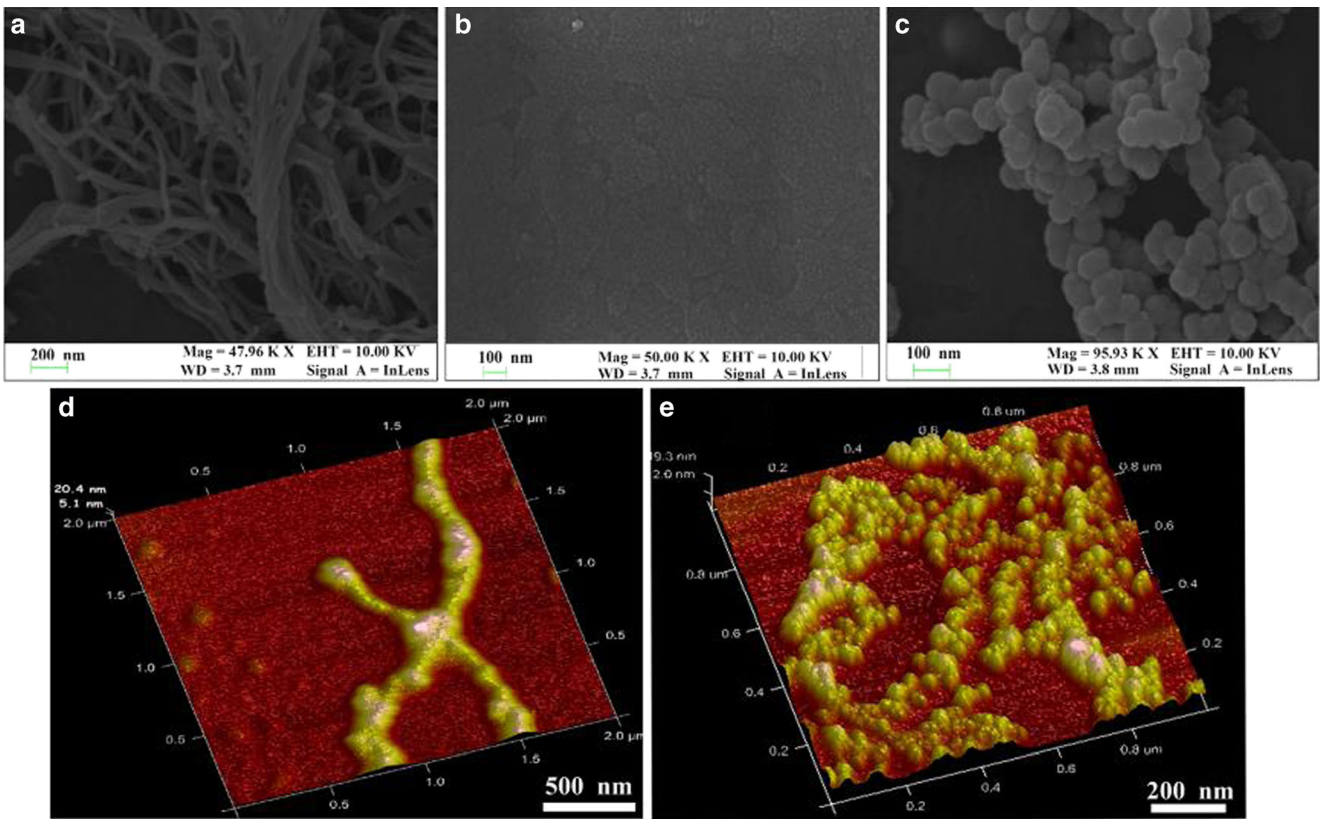


Fig. 1 High resolution scanning electron microscope (HESEM) images of (**a**) PANI-NF film, (**b**) CdS NCs/PANI-NF, (**c**) CdS-Cr/PANI-NF; Atomic force microscope images (**d**) of CdS NCs/PANI-NF, (**e**) CdS-Cr/PANI-NF

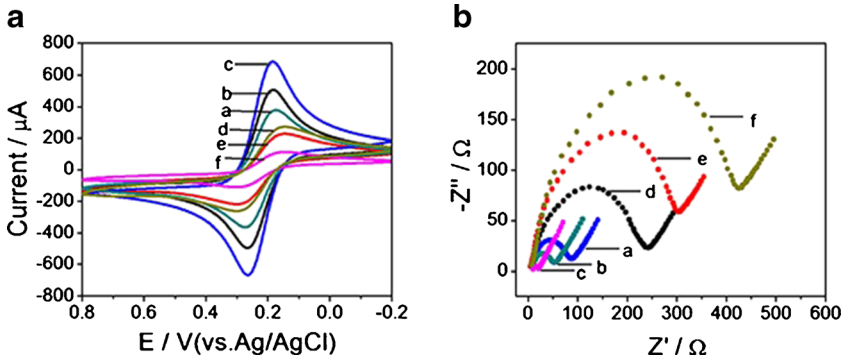


Fig. 2 Results from: (**a**) Cyclic voltammetry, and (**b**) Electrochemical impedance spectroscopy at different modified electrodes in 5 mM $[Fe(CN)_6]^{4-/3-}$(1:1)+0.5 M KCl solution. Curves: (*a*) bare GCE, (*b*) PANI-NF/GCE, (*c*) CdS-Cr/PANI-NF/GCE, (*d*) heparin/CdS-Cr/PANI-NF/GCE, (*e*) bovine serum albumin/heparin/CdS-Cr/PANI-NF/GCE, (*f*) thrombin/bovine serum albumin/heparin/ CdS-Cr/PANI-NF /GCE

the electron transfer and mass transfer of $Fe(CN)_6^{4-/3-}$ at the modified GCE surface.

The electrochemical impedance spectroscopy was shown in Fig. 2b. Nyquist plots comprise a semicircle part at higher frequency range and a straight linear part at lower frequency range. The diameter of the semicircle equals to the electron transfer resistance (R_{et}) at the electrode interface. Due to the good electron conductivity, the PANI-NF film (curve b) and CdS-Cr nanoprobes-modified electrode (curve c) exhibited lower R_{et} than the bare GCE (curve a). It was clear that the diameter of the semicircles increased successively with sequential assembly of heparin (curve d), BSA (curve e) and specific recognition of thrombin (curve f), indicating an enhancement of the R_{et} step by step. This result was consistent with the above cyclic voltammetry observations, and indicated that the biosensor had been successfully prepared.

Electrochemiluminescence behavior of the biosensor

The ECL behavior of the biosensor was investigated in phosphate buffer (0.1 M, pH 7.4) with 0.1 M $K_2S_2O_8$ during the CV scanning. In Fig. 3a, it exhibited ~6-fold enhancement of the ECL intensity as compared to pure CdS NCs under the same conditions. It was attributed to Cr ions loaded mostly on the surface of CdS NCs to form sulfide; and was also incorporated in CdS NCs to build a new surface state-Cr ion complex to improve the luminescence efficiency by filling the surface defects of CdS NCs [16, 18].

The ECL behavior of the modified electrodes was showed in Fig. 3b. The bare GCE and PANI-NF film-modified electrode produced a weak ECL response (curve a and b). However, after modifying CdS-Cr onto the PANI-NF film, a remarkable ECL increase was observed (curve c). When heparin and BSA was immobilized onto the CdS-Cr/PANI-NF modified electrode respectively, the ECL signal decreased apparently (curve d and e) due to the diffusion of coreactant was hindered toward the electrode surface. An interesting phenomenon can be observed in curve f, the ECL intensity had an obvious decreased after incubation with thrombin, which due to the specific recognition of thrombin with heparin. As a result, we made a conclusion that this ECL biosensor can be used to detect thrombin.

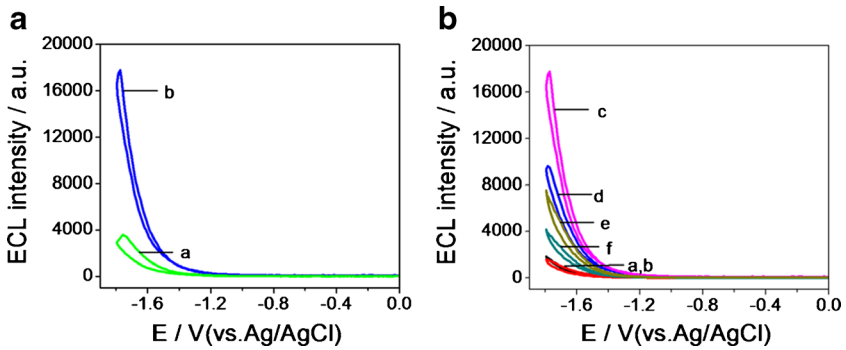


Fig. 3 **a** The comparison of ECL behaviors curves for (*a*) CdS NCs/PANI-NF/GCE, (*b*) chromium-capped CdS nanoprobes (CdS-Cr)/PANI-NF/GCE. **b** ECL behaviors curves at different modified electrodes: (*a*) bare GCE, (*b*) PANI-NF/GCE, (*c*) CdS-Cr/PANI-NF/GCE, (*d*) heparin/CdS-Cr/PANI-NF/GCE, (*e*) bovine serum albumin/heparin/CdS-Cr/PANI-NF/GCE, (*f*) thrombin/bovine serum albumin/heparin/ CdS-Cr/PANI-NF /GCE, in phosphate buffer (0.1 M, pH 7.4) containing 0.1 M $K_2S_2O_8$

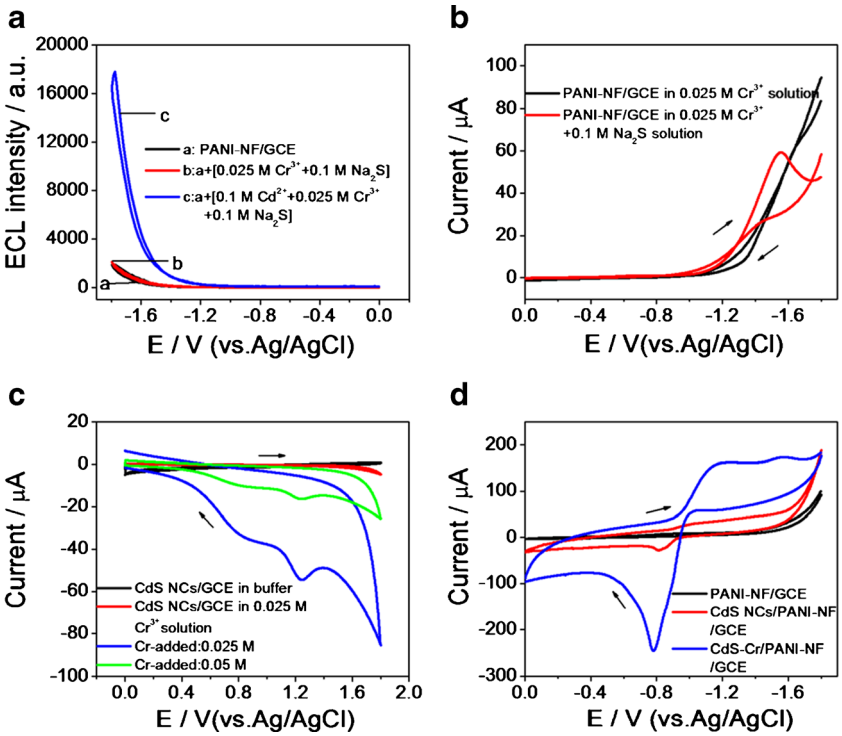


Fig. 4 **a** ECL behaviors curves for (*a*) PANI-NF/GCE, (*b*) PANI-NF/GCE in solution containing 0.025 M Cr^{3+} and 0.1 M Na_2S, (*c*) CdS-Cr/PANI-NF/GCE; Cyclic voltammograms of **b** PANI-NF/GCE in N_2-saturated phosphate buffer (0.1 M, pH 7.4) containing 0.025 M Cr^{3+},0.025 M Cr^{3+} + 0.1 M Na_2S, respectively; **c** CdS NCs/PANI-NF/GCE, CdS NCs/PANI-NF/ GCE+0.025 M Cr^{3+} ions, 0.025 M CdS-Cr and 0.05 M CdS-Cr modified electrode respectively, in N_2-saturated phosphate buffer (0.1 M, pH 7.4), Scan range: 0 – 1.8 V. Scan rate: 100 $mV \cdot s^{-1}$; (**d**) PANI-NF/GCE, CdS NCs/PANI-NF/GCE, CdS-Cr/PANI-NF/GCE in N_2-saturated phosphate buffer (0.1 M, pH 7.4)

Electrochemiluminescence mechanism of CdS-Cr nanoprobes

In order to verify the ECL mechanism of CdS-Cr nanoprobes, a series of experiment was implemented. As shown in Fig. 4a, PANI-NF film modified electrode produced a weak ECL response. When the modified electrode by PANI-NF was dipped into the solution containing 0.025 M Cr^{3+} and 0.1 M Na_2S via in-situ electrochemical synthesis, the electrode produced a weak ECL response indicating Cr^{3+}-sulfide produced ECL response. However, the CdS-Cr nanoprobes produced a strong ECL response. The electrochemical behaviors of Cr^{3+} and Cr^{3+}-sulfide were also investigated under cyclic voltammetric scanning in the negative direction from 0 to −1.8 V. The modified electrode by PANI-NF film produced a reduction peak (−1.6 V) in the solutions containing Cr^{3+} and Na_2S, and whereas, the peaks were not be observed in the solutions only containing Cr^{3+} (Fig. 4b). Also two reduction peaks (1.2 and 0.61 V) of CdS-Cr nanoprobes were clearly observed in Fig. 4c. However, there were no such peaks in the CdS NCs solution or CdS NCs-Cr ions mixed solution. Thus, the reduction was not from CdS NCs or Cr^{3+} ions, or Cr^{3+}-adsorbed CdS NCs, but from the intermediate states caused by Cr^{3+} loading [28]. With a loading level raised to 0.05 M, reduction peaks became less obvious due to Cr-Cr interactions, which was well agreement with ECL intensity discussions.

The ECL emission performance of CdS-Cr nanoprobes was closely related to the stability of the reduced CdS-Cr. Figure 4d showed the electrochemical behaviors of CdS-Cr nanoprobes. For pure CdS NCs, a pair of oxidation and reduction peaks (−0.8 and −1.1 V) was observed. The reduced-state CdS-Cr nanoprobes (−1.1 and −1.6 V) were formed in the negative direction scan were so stable, and they were oxidized again (−0.8 V) when scanning back without the presence of

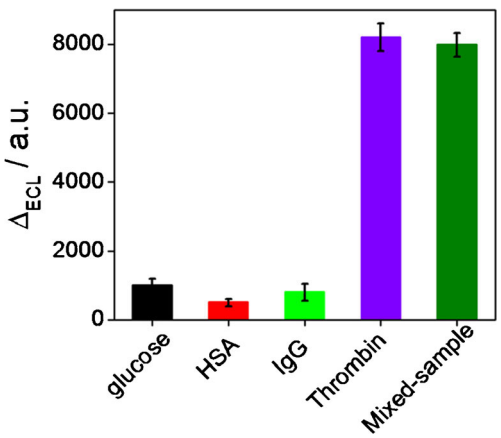


Fig. 5 ECL signal changes of the ECL biosensor in a 1.0 pM glucose, 1.0 pM human serum albumin (HSA), 1.0 pM human IgG, 1.0 pM thrombin. The mixed-sample contained 1.0 pM thrombin, 10.0 pM glucose, 10.0 pM HSA and 10.0 pM IgG, in phosphate buffer (0.1 M, pH 7.4) containing 0.1 M $K_2S_2O_8$, scan rate: 100 $mV \cdot s^{-1}$

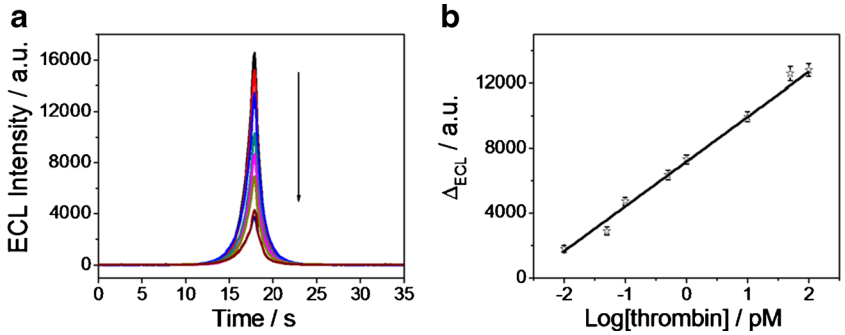


Fig. 6 **a** ECL signal curves of the modified biosensor in the presence of different concentrations of thrombin in phosphate buffer (0.1 M, pH 7.4) containing 0.1 M $K_2S_2O_8$, scan rate: 100 $mV \cdot s^{-1}$. Thrombin concentration (pM): 0, 0.01, 0.05, 0.5, 1.0, 10.0, 50.0 and 100.0. **b** Logarithmic calibration curve for thrombin determination, three measurements for each point

a co-reactant, indicating that the reduction was from the intermediate states caused by Cr^{3+} loading. Thus, in the presence of co-reactant $K_2S_2O_8$, the ECL processes of the CdS-Cr nanoprobes were as follows [29, 30]: as the electrode potential became sufficiently negative, the CdS-Cr was reduced to CdS-$Cr^{-\bullet}$, and the coreactant $S_2O_8^{2-}$ was reduced to $SO_4^{-\bullet}$. $SO_4^{-\bullet}$ could react with CdS-$Cr^{-\bullet}$ to obtain an excited state (CdS-Cr*). This state emitted light in the aqueous solution to produce an ECL signal.

$$CdS{-}Cr + ne^- \rightarrow n(CdS{-}Cr) \quad (1)$$

$$S_2O_8^{2-} + e^- \rightarrow SO_4^{2-} + SO_4^{-\cdot} \quad (2)$$

$$(CdS{-}Cr)^{-\cdot} + SO_4^{-\cdot} \rightarrow (CdS{-}Cr)^* + SO_4^{2-} \quad (3)$$

$$(CdS{-}Cr)^* \rightarrow CdS{-}Cr + h\upsilon \quad (4)$$

Optimization of methods

To obtain an optimum interface for sensing, the electrochemical synthesis conditions for CdS-Cr/PANI-NF, including the loading concentration of chromium, the current and the time were optimized. In addition, to apply the proposed biosensor in the ECL assay, several experimental parameters, including scan rate, concentration of heparin and the incubation time of thrombin, should be optimized. The results were given in the Electronic Supporting Material (Fig. S1). It demonstrated that the following experimental conditions were the optimal synthesis conditions: the chromium-loading concentration was 0.025 M, current and time of electrochemical synthesis condition of CdS-Cr on the PANI-NF film was 320 μA and 300 s, respectively. Further, the scan rate of 100 $mV \cdot s^{-1}$, the heparin concentration of 0.8 $mg \cdot mL^{-1}$ and the incubation time of 120 min were selected as the optimal experimental parameters.

Selectivity of the ECL biosensor for thrombin detection

The selectivity of ECL biosensor for thrombin detection was tested via comparing the changes of ECL signal brought by three other proteins: glucose, human serum albumin (HSA), and IgG. In Fig. 5, the ECL responses of the biosensor were compared after incubation with glucose, HSA, and IgG solutions under the same experimental conditions. A slight decrease was observed in the cases of glucose, HSA and IgG. As for the mixed sample (1.0 pM thrombin coexisted with 10.0 pM glucose, 10.0 pM HSA, and 10.0 pM IgG), only slight decrease was found in comparison with only thrombin, suggesting that the ECL biosensor had quite an affinity toward thrombin and was well selective.

Table 1 Performance comparison of different material and method-based biosensors for the analysis of thrombin

Different material and method-based biosensor	Linear range	Detection limit (pM)	Ref.
1. Fe_3O_4@CdSe composite QDs-based ECL aptasensor	1.0 pM–5.0 nM	0.12	[23]
2. Luminal-based ECL aptasensor	0.005 nM–50 nM	1.70	[24]
3. CdTe QDs-based ECL aptasensor	0.5 pM–800 pM	0.35	[31]
4. $Ru(bpy)_3^{2+}$-based ECL aptasensor	0.90 pM–226 pM	0.40	[32]
5. Ferrocene-AuNPs-based differential pulse voltammetry aptasensor	12 pM–1.2 μM	100	[33]
6. In-situ generated CdS-Cr nanoprobes and heparin-based ECL biosensor	0.01 pM–100 pM	0.0068	This work

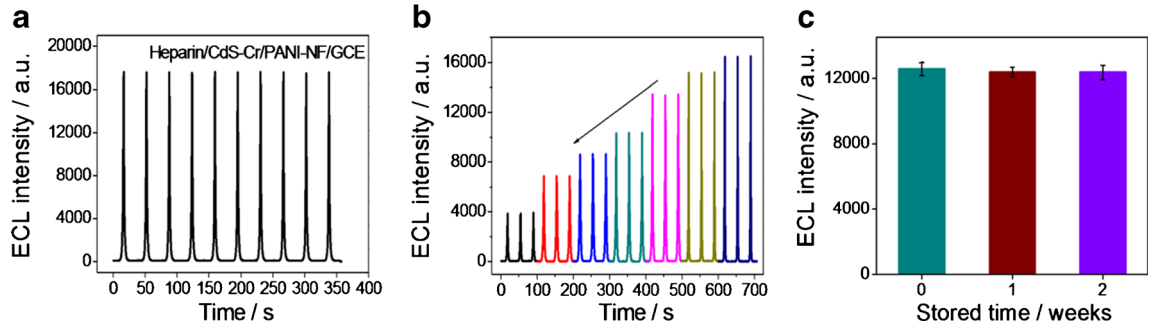


Fig. 7 **a** ECL behaviors curves of heparin/ CdS-Cr/PANI-NF/GCE in phosphate buffer (0.1 M, pH 7.4) containing 0.1 M $K_2S_2O_8$ under continuous cyclic potential scan for 10 cycles from 0 to −1.8 V (vs.SCE), scan rate: 100 mV·s^{-1}. **b** ECL signal curves of consecutive cyclic potential scans to various concentrations of thrombin (pM): 0, 0.01, 0.05, 0.5, 1.0, 10.0, and 100.0. **c** ECL signal response of the biosensor for 0, 1 and 2 weeks, respectively (*error bars*: SD, n=3)

Analytical performance of the modified biosensor

Under the optimum conditions, quantitative analysis of thrombin was performed on the biosensor by using the electrochemiluminescence technique. The ECL intensity decreased with an increase of the concentration of thrombin, as shown in Fig. 6a. The Δ_{ECL} was found to be logarithmically related to the concentration of thrombin in the range from 10 fM to 100 pM. The regression equation was Δ_{ECL}= 3490+1574 log C (the unit of C is pM) with a correlation coefficient of 0.9984. Where C was the concentration of thrombin; Δ_{ECL} was the relative ECL intensity calculated by I_0 - I, where I_0 and I were the ECL intensity without and with thrombin, respectively (Fig. 6b). The detection limit for thrombin was calculated to be 6.8 fM at 3σ. In this context, this method performed better than several other published methods in term of linear range and detection limit (Table 1); it may be attributed to that the technique for in-situ electrochemical preparation of the nanoprobe yielded a sensor surface with favorable space structure, positive charge, stability and amplified the signal of ECL intensity; also the thrombin-heparin interaction served as a new and better recognition tool for thrombin.

Table 2 Recovery of thrombin assay at different concentrations spiked into human serum samples[a]

Sample[b]	Spiked (pM)	Found (pM)	Recovery (%)	RSD (%)
Serum sample 1	5.0	4.6	92.0	4.4
	10.0	10.7	107.0	5.0
	100.0	103.4	103.4	6.2
Serum sample 2	5.0	4.5	90.0	5.8
	10.0	10.2	102.0	6.3
	100.0	104.8	104.8	6.1

[a] Each sample was repeated for three times and averaged to obtain the recovery and RSD values

[b] All human serum samples were diluted 10-fold prior to assay

Stability and reproducibility of the modified biosensor

The stability of the modified electrode by heparin/CdS-Cr/PANI-NF was studied by the continuous potential scanning for 10 cycles, and the ECL signal-time curve was shown in Fig. 7a.

The stability of the biosensor was evaluated under consecutive cyclic potential scans to various concentrations of thrombin (Fig. 7b). It showed that the ECL intensity decreased with the increasing concentration of thrombin, and a relative stable curve at each concentration could be obtained. Also, the stability of the biosensor was studied by the discontinuous measurement of the ECL signal response for 1.0 pM thrombin. After the biosensor was stored in phosphate buffer (0.1 M, pH 7.4) in the refrigerator at 4 °C for 1 and 2 week; about 97.8 % of the original ECL response was retained after 1 week, and about 94.2 % of the initial ECL signal was noticed after 2 weeks, respectively (Fig. 7c). The reproducibility of the proposed ECL biosensor was investigated with interassay. The interassay variation coefficient was 4.53 %, which was evaluated from the response to 1.0 pM thrombin at six different electrodes independently. Thus, this fact demonstrated the proposed ECL biosensor possessed acceptable reproducibility.

Serum sample analysis of thrombin

The potential applicability of the ECL biosensor was investigated by determining spiked thrombin in human blood serum [34]. In the experiment, we spiked thrombin into 10-fold-diluted serum samples to examine the applicability of this ECL method in real serum sample. The serum samples were spiked with 5.0, 10.0, 100.0 pM, and it was measured by our ECL biosensor. The results showed that the recoveries varied from 90.0 to 107.0 % by comparing the measured amounts to that of added thrombin (Table 2). Thus it suggested that this ECL biosensor has a promising potential application in real biological samples.

Conclusions

We successfully developed a simple and sensitive ECL biosensor with the use of a GCE electrode modified with the film of electro-polymerized polyaniline nanofibers, which was followed by in-situ synthesized CdS-Cr nanoprobes on the film. CdS-Cr nanoprobes amplified the signal of ECL intensity and improved the detection sensitivity. Interesting, it showed about 6-fold enhancement of ECL intensity compared to pure CdS NCs when synthesized in aqueous solution system. The thrombin-heparin interaction served as a new recognition tool for thrombin and represented an attractive alternative to respective aptamers. This ECL biosensor exhibited a wide dynamic range from 10 fM to 100 pM for thrombin, and a lower detection limit of 6.8 fM at 3σ. The technique for in-situ electrochemical synthesis of the nanoprobe is simple, facile, and yields a sensor surface with favorable space structure, positive charge and stability. Moreover, this method opens a promising path for other NCs to improve their ECL behaviors, and can be further applied to develop highly sensitive ECL sensing systems.

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